Does the AI Time Horizon Predict Anything? A Preregistered Predictive-Validity Shootout

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Abstract

METR's 50% time horizon—the human-task duration an AI completes with 50% success—is the most quoted single number in AI capability evaluation. Recent work improved how the number is *estimated*, but nobody has tested whether it *predicts* anything. We preregistered a three-way predictive shootout on METR's public Time Horizon 1.1 data (23,235 runs, 228 tasks, 20 AIs): does a model's spline-estimated success curve predict held-out task success better than (a) its mean accuracy and (b) an IRT ability estimate, under 5-fold cross-validation over task families with strictly proper scoring rules? Both preregistered hypotheses are confirmed: the horizon predictor beats mean accuracy (paired log-loss difference -0.238 ± 0.033) and the plain-1PL IRT-ability implementation (-0.418 ± 0.122). Two post-hoc qualifications temper the story: the spline is statistically tied with a plain shared-slope logistic regression ($+0.004 \pm 0.008$, n.s.), so the gain comes from using human-time information at all rather than from the spline nonlinearity; and the IRT-ability predictor, while rank-competitive (AUC 0.905 ± 0.024 vs 0.919 ± 0.015), is badly miscalibrated (expected calibration error 0.155 vs 0.031), scoring worse than the constant mean on 3 of 5 folds. The horizon number is an interpretable repackaging of the success curve, not predictive content beyond the curve itself.

1 Introduction

When a new AI model is released, one number now routinely anchors the capability discussion: its *time horizon*, the duration of human-expert work the model can complete with 50% reliability (Kwa et al., 2025). Estimated by fitting a success curve over tasks of known human duration, the 50% horizon has doubled roughly every seven months since 2019, and the plot of horizons over time is a widely reproduced figure in AI forecasting.

A recent paper by Nguyen & Fithian (2026) substantially improved the *estimation* of these curves, replacing METR's log-linear logistic specification with monotone splines and explanatory item-response-theory (IRT) models, and validating the improvement under cross-validated proper scoring rules. What remains untested is the *predictive validity* of the metric itself: does the horizon machinery—a per-model success curve anchored to human time—actually predict anything better than the cheap summaries any practitioner already has? Concretely: given a new task of known human duration, is a model's predicted success from its horizon curve better than just guessing the model's average accuracy, or than an IRT ability estimate?

We answer with a preregistered shootout. Our contributions:

1. A locked analysis plan (§3) written *before* any model was fit, specifying three predictors, a 5-fold CV scheme over task families, proper scoring rules, and a mechanical decision criterion.

2. Confirmation of both preregistered hypotheses on METR's public data: the spline-horizon predictor beats mean accuracy and IRT ability on held-out log-loss, by -0.238 ± 0.033 and -0.418 ± 0.122 respectively.

3. Two honest qualifications from clearly labeled post-hoc analyses: the spline ties a plain shared-slope logistic (the predictive content is the human-time covariate, not the nonlinearity), and the IRT predictor is rank-competitive but catastrophically miscalibrated.

2 Background

2.1 Time horizons

Kwa et al. (2025) define the q-task-completion time horizon of an AI as the human time t such that the model's success probability $p_j(t)$ equals q; $q = 0.5$ is the headline number. Tasks come with T_i, the minutes a human expert needs (measured or expert-estimated), and each AI's curve $p_j(t)$ is fit by logistic regression of binary outcomes on $\log_2 T_i$. We use METR's public Time Horizon 1.1 release: 24,008 runs over 228 tasks in 79 families (HCAST, SWAA, RE-Bench), 20 AIs plus a human baseline (METR, 2026).

2.2 Improved estimation

Nguyen & Fithian (2026) observe that log-linearity is a strong and apparently false assumption—their diagnostics show difficulty is roughly flat over human times of 2–30 minutes—and propose two replacements. *Model 1* fits $\operatorname{logit} p_j(t) = \alpha_j - \beta\, f(\log_2 t)$ with f a shared monotone I-spline (4 interior knots). *Model 2* is an explanatory IRT model: a Rasch form with shared slope, family and idiosyncratic random effects, and beta-binomial overdispersion, whose latent difficulties are regressed on $\log_2 T_i$. Both beat the log-linear baseline under 5-fold CV with proper scoring rules. Their focus is estimation; ours is whether the resulting numbers predict.

2.3 Proper scoring rules

We evaluate probabilistic predictions with strictly proper scoring rules, which uniquely reward honest probability forecasts (Gneiting & Raftery, 2007). Primary: the logarithmic score (log-loss). Secondary: the Brier (quadratic) score and AUC. We follow Nguyen & Fithian (2026) in using METR's sqrt-family weights ($1/\sqrt{n_F}$ per task family, normalized per AI) for fitting and scoring.

3 Preregistered shootout

The full locked plan (2026-10-11, before any fit) is summarized in Appendix A. In brief:

Data. All AI runs from the public release above (23,235 runs; the human alias excluded). Outcome $Y_{ijr} \in \{0, 1\}$; covariate $x_i = \log_2 T_i$; weights w_{ijr} as shipped.

Predictors. P1 (horizon/spline): $\operatorname{logit} p_j(t) = \alpha_j - \beta f(\log_2 t)$, $\beta > 0$, f monotone cubic (PCHIP through strictly increasing knot values), 4 interior knots at train quantiles, $f(\min) = \min$, $f(\max) = \max$; fit by weighted binary cross-entropy (L-BFGS-B). The 50% horizon is $2^{f^{-1}(\alpha_j/\beta)}$. **P2 (mean accuracy):** per-AI weighted train mean, constant on held-out tasks. **P3 (IRT ability):** 1PL Rasch $\operatorname{logit} p_{ij} = \alpha_j - \beta\theta_i$ on train tasks, then OLS $\theta_i = a + bx_i$; held-out tasks use $\hat{\theta}_i = a + bx_i$. (We noted in the preregistration that P3 approximately reduces to a shared-slope logistic in $\log_2 T$ for unseen tasks.)

CV and decision rule. 5-fold CV over the 79 task families (fixed seed; all runs of a task in one fold). Primary metric: weighted log-loss on held-out runs. P1 "wins" iff its mean paired fold-difference vs P2 is negative and vs P3 is negative, each with $|\text{mean}| > 2\,\text{SE}$ (SE of the 5 paired differences). H1/H2 are the P1-vs-P2 and P1-vs-P3 wins. Retrospectively, the $|\text{mean}| > 2\,\text{SE}$ bar over $n = 5$ folds corresponds to a minimum detectable standardized effect of $d_z \approx 1.3$ at 80% power; both confirmed hypotheses clear it (|mean|/SE of 7.2 and 3.4).

Table 1: Cross-validated held-out scores, mean ± SE over 5 task-family folds. Lower is better for log-loss and Brier; higher for AUC. P0 is a post-hoc reference (log-loss only).

Predictor	Log-loss	Brier	AUC
P1 spline / horizon	$\mathbf{0.394 \pm 0.018}$	$\mathbf{0.128 \pm 0.007}$	$\mathbf{0.919 \pm 0.015}$
P0 logistic (post-hoc)	0.390 ± 0.025	—	—
P2 mean accuracy	0.632 ± 0.019	0.220 ± 0.009	0.679 ± 0.040
P3 IRT ability	0.812 ± 0.135	0.178 ± 0.028	0.905 ± 0.024

4 Results

4.1 Main comparison

Table 1 reports cross-validated scores. Both preregistered hypotheses are confirmed:

- **H1:** P1 − P2 log-loss $= -0.238 \pm 0.033$ ($|\text{mean}| > 2\,\text{SE}$). The horizon predictor beats mean accuracy on all 5 folds.
- **H2:** P1 − P3 log-loss $= -0.418 \pm 0.122$ ($|\text{mean}| > 2\,\text{SE}$). The horizon predictor beats the IRT predictor on all 5 folds.

The locked rule uses 2 SE as a mechanical bar; with $n = 5$ folds, the two-sided 95% t critical value (4 df) is 2.78, and both wins survive it comfortably: 95% t-CIs are $(-0.331, -0.145)$ for H1 and $(-0.757, -0.080)$ for H2. (Fold test sizes vary from 3,393 to 5,916 runs; the preregistered rule averages folds unweighted, which we retain as locked.)

Figure 1 shows the per-fold pattern: P1 dominates P2 and P3 in every fold, while P3's log-loss explodes in two of the five folds (0.92, 1.27), exceeding even the constant mean. The secondary metrics complicate the story: by Brier, the point ordering of P2 and P3 flips (P3 0.178 ± 0.028 is lower than P2 0.220 ± 0.009; paired difference -0.042 ± 0.036, n.s.). Brier punishes extreme miscalibrated probabilities far less than log-loss does, so this flip is consistent with the "rank-good, probability-bad" diagnosis below rather than a reprieve for P3. For AUC, P1 beats P2 clearly ($+0.240 \pm 0.054$) but is statistically tied with P3 ($+0.014 \pm 0.010$): IRT ability *ranks* held-out tasks about as well as the spline, yet produces far worse probabilities.

4.2 Horizon estimates

Figure 2 compares our spline 50% horizons (fold mean ± sd) against METR's published logistic estimates. The ordering of models is preserved, but individual values shift substantially (ratios 0.3–2.5), consistent with Nguyen & Fithian (2026)'s finding that relaxing log-linearity moves horizons outside the original confidence intervals. Uncertainty is large for frontier models (e.g. Claude Opus 4.6: 689 ± 629 min across folds). The fitted time-to-difficulty curve (Figure 4, Appendix) reproduces the reported flat region over roughly 2–30 human minutes.

The number is noisy; the curve is not. The per-fold instability of the 50% point estimates stands in sharp contrast to the stability of the underlying predictive exercise: the same folds on which horizons swing by factors of two produce nearly identical held-out log-losses (P1 fold range 0.336–0.439). The scalar horizon—the most quoted number—is a fragile functional of the curve (a single level crossing), while the curve itself predicts stably. The number that gets quoted is the least stable part of the analysis; the horizon earns its place on the plot as an anchor, not an oracle.

5 Post-hoc analyses (not preregistered)

A plain logistic ties the spline. We added **P0**, a shared-slope logistic $\text{logit}\, p_j = \alpha_j - \beta x$ fit under the identical CV: mean log-loss 0.390 ± 0.025 vs P1's 0.394 ± 0.018 (paired diff $+0.004 \pm 0.008$, n.s.; 95% t-CI

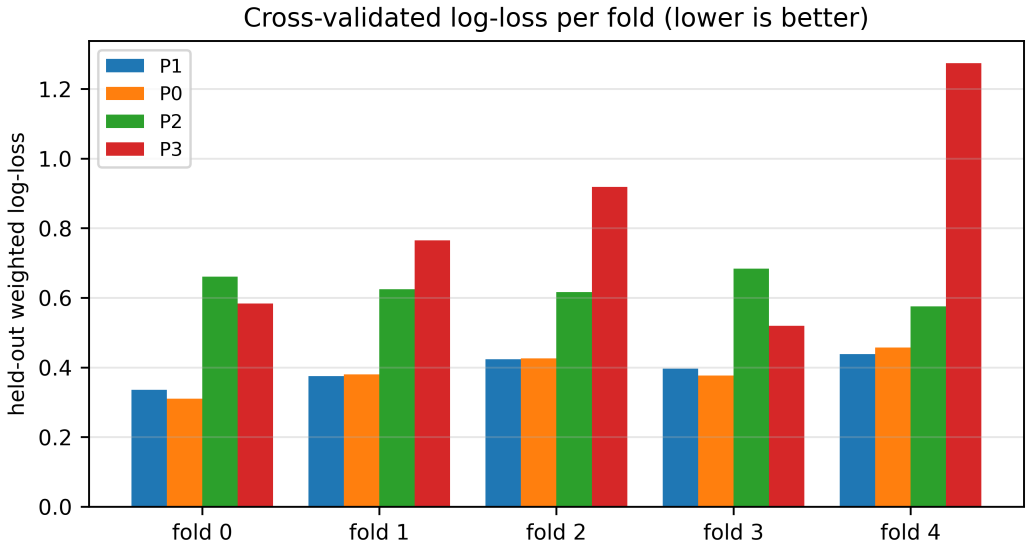


Figure 1: Held-out weighted log-loss per fold (lower is better). P1 and the post-hoc logistic P0 dominate P2 and P3 in every fold; P3 is worse than the constant mean in 3 of the 5 folds.

$(-0.019, +0.026)$), and the same tie under equal-task weighting (0.347 vs 0.349). The interval rules out any advantage comparable in scale to the -0.238 P1-vs-P2 effect on this data. On this 20-AI public slice, the spline's nonlinearity adds no significant predictive content beyond the linear model—the win over the baselines comes from conditioning on human time at all. (As with all post-hoc comparisons, this one was chosen after seeing the data and should be read with the usual garden-of-forking-paths caution: post-hoc comparison choices can inflate apparent effects.) This partially qualifies Nguyen & Fithian (2026)'s CV edge for the spline, which was established on 26 AIs and a 12-metric suite; our narrower replication on the public subset does not reproduce it at significance.

IRT ability is rank-good, probability-bad. The reliability diagram (Figure 3) shows P3 systematically overconfident: tasks assigned 15–35% success actually succeed 51–61% of the time. Expected calibration error (10 equal-width bins on predicted probability, pooled out-of-fold predictions, METR sqrt-family weights within and across bins): P3 0.155 vs P1 0.031, P2 0.037, P0 0.024. The mechanism is now diagnosed, not asserted (per-fold fits in Appendix D): the Rasch slope fits steep in every fold ($\hat{\beta} = 6.8$–12.1 on a latent scale of sd ≈ 1.1–1.7), which is what the training data wants, but the unpenalized OLS extrapolation of difficulties to unseen tasks feeds it $\hat{\theta}$ values whose modest errors it amplifies into logits of ± 10–20: 46–85% of held-out predictions land below 0.01 or above 0.99 (probabilities were clipped to $[10^{-12}, 1 - 10^{-12}]$ per the locked plan, so these extremes are genuine, not a scoring artifact). Our preregistered expectation—that P3 would "approximately reduce to a shared-slope logistic"—failed precisely here: the jointly estimated logistic (P0) has no two-stage compounding of noisy-$\hat{\theta} \to$ OLS $\to$ extrapolation, so it never reaches for these extremes. A ridge- penalized variant did not help ($\hat{\beta}$ simply grew to compensate), confirming the steepness is load-bearing for the fit rather than removable by naive shrinkage. IRT ability without an overdispersion component (cf. Model 2 of Nguyen & Fithian (2026)) is a respectable ranker but a dangerous probability forecaster for unseen tasks.

Sensitivity. Rescoring all predictors with equal-task weights ($1/n_{ij}$ per run) preserves the ordering (P1 0.349, P0 0.347, P2 0.629, P3 0.723). Per-AI paired differences (Appendix C) show P1's gain over P2 is largest for mid-capability models, while its gain over P3 concentrates in frontier models where IRT extrapolation is most extreme.

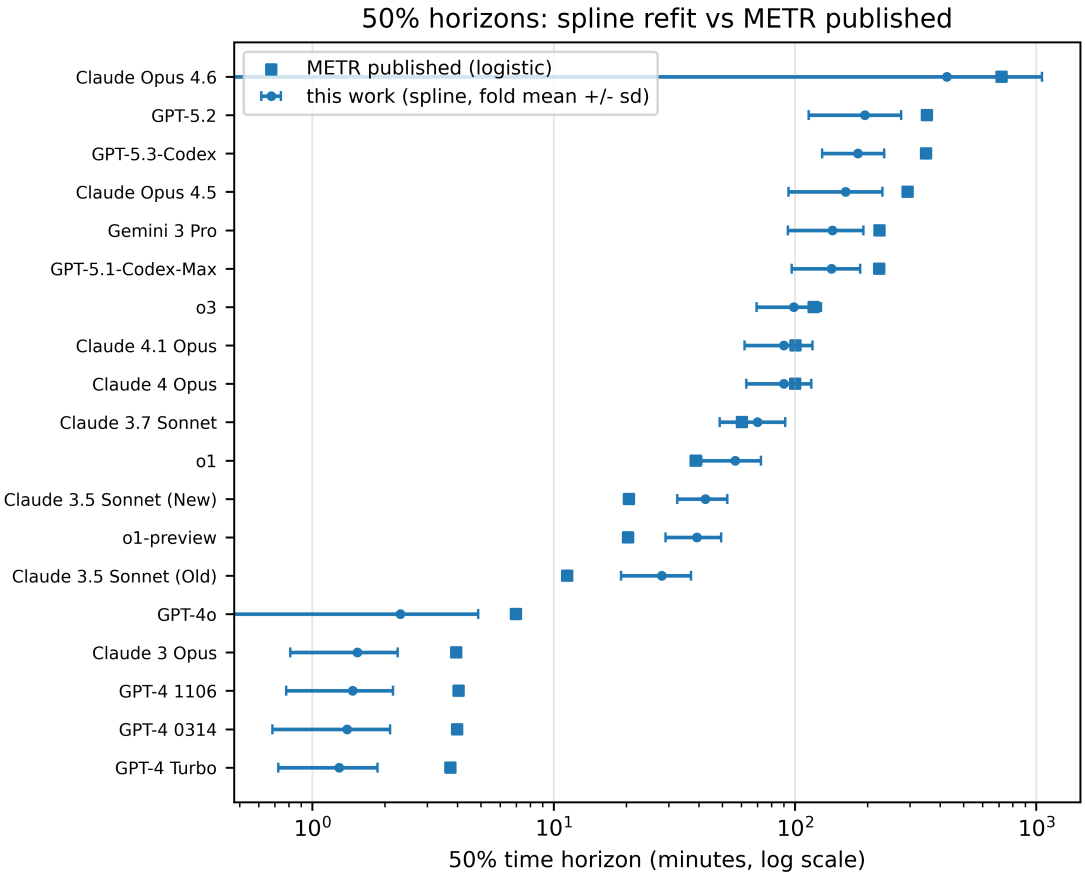


Figure 2: 50% time horizons: this work's spline refit (per-fold mean ± sd) vs METR's published logistic estimates. Model ordering is stable; point values shift, with large fold-spreads at the frontier.

6 Discussion

What the horizon number is. Our results locate the 50% horizon's value precisely: as a *predictor*, the horizon machinery (P1) beats the obvious cheap baselines, but the predictive content is the human-time covariate, not the spline—a linear model does as well. The horizon *number* is therefore best understood as an interpretable repackaging of the success curve into human-time units, in the spirit of the "capability anchoring" framing of Nguyen & Fithian (2026): useful for communication and forecasting trends, not extra information beyond $p_j(t)$.

A warning on IRT ability, with a caveat on our implementation. Ability estimates from joint IRT fits are tempting portable summaries of models, but our P3 shows they extrapolate poorly as probabilities: good AUC, terrible log-loss. Anyone using IRT abilities to forecast success on new tasks should recalibrate—or use the overdispersed specification of Nguyen & Fithian (2026). We stress, however, that P3 as implemented here—a plain 1PL with an unpenalized OLS extrapolation of difficulties, no family effects, no beta-binomial overdispersion—is not a fair representative of IRT-based forecasting in general; the overdispersion component it lacks is precisely what tempers extreme probabilities. H2 should therefore be read as "the horizon predictor beats this plain-1PL implementation," not as an indictment of IRT ability.

Limitations. The task suite is observational, not a sample of real work; 20 public AIs are a convenience sample (the source paper used 26); 29% of human times are expert estimates rather than measurements (Nguyen & Fithian, 2026); runs within a (task, AI) pair are non-independent, so our SEs use folds (families)

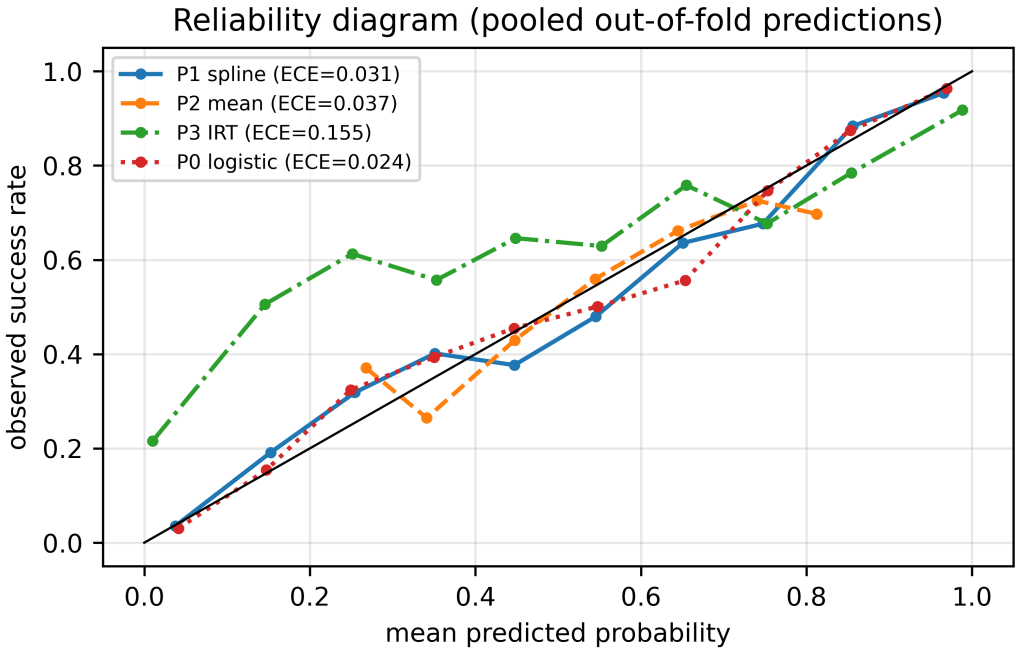


Figure 3: Reliability diagram on pooled out-of-fold predictions. P1, P2, and the post-hoc P0 track the diagonal; P3 (IRT) is strongly miscalibrated, assigning low probabilities to tasks that succeed most of the time.

as the unit, and with only 5 folds the t-based intervals above are wide by construction. Our P3 is a plain 1PL without family effects or beta-binomial overdispersion, which likely explains part of its miscalibration relative to the paper's Model 2; see the caveat above. The analysis uses a single fixed fold partition (seed 20261012); post-hoc seed sensitivity is not shown.

7 Conclusion

We preregistered the shootout and confirmed both hypotheses: AI time horizons predict held-out task success far better than mean accuracy or IRT ability. We qualify that result honestly: the spline is no better than logistic regression on log human time, and IRT ability needs calibration before it forecasts probabilities. The horizon earns its place on the plot— as an anchor, not as an oracle.

Statement of AI Use

The author used the AI tool Muse Spark (version 1.3, Meta) for analysis code, figure generation, manuscript drafting, and grammatical editing. The contents are reviewed, edited, and attested for accuracy by the author.

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A Preregistration summary

The analysis plan was locked on 2026-10-11 before any model was fit (only descriptive data checks—counts, fields, weight sums—preceded it). It specified: the dataset and filters (§3); exact predictor definitions P1/P2/P3 including knot placement, identifiability constraints, and optimizer; 5-fold CV over task families with seed 20261012; weighted log-loss as the primary metric with Brier and AUC secondary; and the mechanical decision rule (|mean paired diff| $>$ 2 SE vs both baselines). P0, calibration, weighting-sensitivity, and per-AI analyses were not preregistered and are labeled post-hoc throughout. No deviations from the locked plan occurred.

B Fitted time-to-difficulty curve

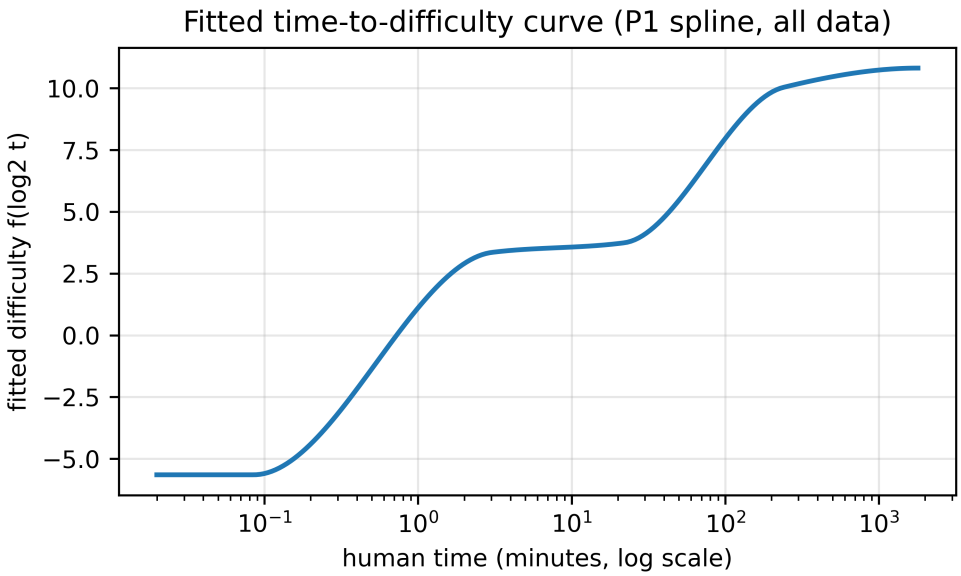


Figure 4: Monotone spline f mapping $\log_2$ human minutes to difficulty (P1 fit on all data). Difficulty is flat over roughly 2–30 minutes, reproducing the qualitative finding of Nguyen & Fithian (2026).

C Per-AI paired differences

Mean paired log-loss difference (P1 minus baseline) by AI, averaged over folds; negative favors P1. Largest P1-vs-P2 gains: GPT-4 Turbo (−0.319), o1-preview (−0.318), GPT-4 0314 (−0.311). Largest P1-vs-P3 gains: Claude Opus 4.6 (−0.717), Claude Opus 4.5 (−0.621), GPT-5.2 (−0.599).

Table 2: Per-fold P3 (1PL + OLS extrapolation) fit diagnostics. $\hat{\beta}$ is the Rasch slope; $\hat{\theta}$ ranges are extrapolated difficulties on held-out tasks; "extreme" is the fraction of held-out predictions below 0.01 or above 0.99.

Fold	$\hat{\beta}$	train θ sd	test $\hat{\theta}$ range	extreme frac.	log-loss
0	8.61	1.13	$[-1.73, 1.31]$	0.85	0.583
1	12.10	1.15	$[-0.48, 1.25]$	0.46	0.765
2	7.01	1.65	$[-0.33, 1.92]$	0.50	0.919
3	6.76	1.40	$[-2.40, 1.91]$	0.76	0.519
4	7.69	1.44	$[-1.38, 1.70]$	0.67	1.274

D P3 per-fold diagnostics

Data and Code Availability

The analysis code, per-fold results, and the one-command reproduction package (`bash reproduce.sh`, 39/39 checks) are included as supplementary material. The METR time-horizon dataset used for the shootout is publicly released by METR. A public code repository will be released upon acceptance.

Competing Interests

The author declares no competing interests. No funding was received for this work.